Exploratory Data Analysis: Public University Funding and the National Tax Take

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Overview and Research Question

This exploratory data analysis (EDA) inspects the newly constructed global fiscal and education panel (`data/processed/clean_tax_panel.csv`, spanning 1,736 country-waves across 217 sovereign countries from `1985-1989` to `2020-2024`). Our central question is whether increased national spending on university (tertiary) education produces short-term (+5-year) and long-term (+10-year) benefits to a country's tax take once mediating macroeconomic variables are separated out.

To avoid the mechanical identities and ratio distortions identified in previous macro-education studies, this dataset implements three structural improvements:

1. **Four Pure Disaggregated Fiscal Outcomes (Zero GDP Product Identities):**
   - Y_1: **Total Domestic Non-Resource Tax-to-GDP Change (Δ Tax/GDP, percentage points of GDP)** (alongside pure unadjusted World Bank WDI Δ Tax/GDP).
   - Y_2: **Direct Income, Profit, and Capital Gains Tax-to-GDP Change (Δ IncomeTax/GDP, percentage points of GDP)**.
   - Y_3: **Direct Income and Profit Share of Revenue Change (Δ IncomeTax/Rev, percentage points of revenue)**.
   - Y_4: **Indirect Goods and Services (Consumption / VAT) Share of Revenue Change (Δ GoodsTax/Rev, percentage points of revenue)**, which acts as a built-in specificity (negative-control) channel.
2. **Unbundling the UNESCO Denominator Trap:** We compare three strictly observed university funding exposures side by side (with zero cross-sectional proxy imputation):
   - **UNESCO Per-Student Spending Ratio (`tert_spend_per_student_winsor`):** Public spending per university student as a percentage of GDP per person.
   - **Real 2021 PPP Dollars per Student (`real_tert_spend_per_student_ppp`):** Removes GDP per person from the treatment denominator.
   - **Public University Budget Share of GDP (`tert_spend_gdp_pct`):** Removes student headcount from the treatment denominator (paired with total public education spending as a percentage of GDP, `edu_spend_total_gdp_pct`).
3. **Separating Pre-Treatment Controls from Macroeconomic Mediators:** Baseline controls use lagged ($t-1$) GDP per person growth (`lag_gdp_pc_growth_pct`) rather than contemporaneous wave-t growth, while contemporaneous forward GDP growth, enrolment expansion,

inequality changes, and life expectancy changes are reserved explicitly for the **Mediator-Removed Direct Fiscal Effect** model.

Unbundling the UNESCO Denominator Trap Across Income Groups

Table 1 summarises the three university funding measures and baseline tax structure across the four World Bank income groups.

Table 1: University spending indicators and baseline fiscal structure by World Bank income group (medians for skewed spending/enrolment indicators; means for tax shares).

Income Group	Countries (Waves)	UNESCO Ratio (% GDP/person)	**Real PPP $ / Student**	**University Budget (% GDP)**	University Enrolment (%)	Direct Income Tax Share (%)
Low income	21 (80)	**150.0%**	$3,957	0.47%	2.3%	29.2%
Lower middle income	41 (172)	76.5%	$3,258	0.65%	7.4%	30.8%
Upper middle income	54 (219)	26.8%	$3,521	0.64%	25.5%	33.1%
High income	69 (337)	26.9%	**$11,148**	**0.94%**	**58.3%**	**38.4%**

Table 1 illustrates why comparing the raw UNESCO ratio across income groups is misleading without conditioning on income group or unbundling the denominators:

- In **Low income** countries, only **2.3%** of the age cohort attends university, so dividing modest public university budgets (`0.47%` of GDP) by a tiny student headcount and a low GDP per person yields a median UNESCO ratio of **150.0%**.
- Once we unbundle the ratio into **Real 2021 PPP Dollars per Student** and **Public University Budget (% of GDP)**, **High income** countries invest nearly three times as much in real terms (**$11,148** per student vs **$3,957**) and twice as much as a share of national income (**0.94%** of GDP vs **0.47%**), while collecting a substantially higher share of revenue from direct income and profit taxes (**38.4%** vs **29.2%**).

Unadjusted +5-Year and +10-Year Changes in the Four Tax Channels

Table 2 compares the raw mean changes in our four fiscal outcomes between countries investing above their income-group benchmark ($Z = 1$) and those below it ($Z = 0$) across the outlier-screened four-outcome fiscal sample ($N = 372$ country-waves across 121 countries at $+5$ years; $N = 283$ country-waves across 112 countries at $+10$ years, after screening out two raw WDI reporting anomalies in Sudan `1995-1999` and Timor-Leste `2010-2014` where recorded tax revenue exceeded `100%` of GDP).

Table 2: Unadjusted mean changes across the four disaggregated fiscal outcomes by within-income-group university funding regime (`Z_regime_inc`).

University Funding Regime	Horizon	N (Countries)	**Δ Non-Resource Tax / GDP (pp)**	**Δ Direct Income Tax / GDP (pp)**	**Δ Direct Income Tax Share (pp)**	Δ Goods & Services Tax Share (pp)
Low Funding ($Z = 0$)	$+5$ y	203 (83)	`+0.39`	`+0.27`	`+0.90`	`+0.70`
High Funding ($Z = 1$)	$+5$ y	169 (70)	**`+0.33`**	**`+0.40`**	**`+1.76`**	`+0.33`
Low Funding ($Z = 0$)	$+10$ y	148 (73)	`+0.65`	`+0.45`	`+1.53`	`+1.31`
High Funding ($Z = 1$)	$+10$ y	135 (65)	**`+0.74`**	**`+0.65`**	**`+2.64`**	`+0.62`

Even before causal forest adjustment, three patterns stand out in Table 2:

1. **Long-Horizon Tax-Take Acceleration:** While unadjusted $+5$-year changes in overall non-resource tax-to-GDP are similar across groups (`+0.33` vs `+0.39` percentage points), by $+10$ years high university investors ($Z = 1$) expand their non-resource tax-to-GDP ratio by **`+0.74` percentage points** compared with `+0.65` percentage points among low investors ($Z = 0$).
2. **Direct Income and Profit Taxes Drive the Gain:** Direct income, profit, and capital gains taxes as a share of GDP rise by **`+0.40` pp** at $+5$ years and **`+0.65` pp** at $+10$ years in high-investment regimes (versus `+0.27` and `+0.45` pp in low-investment regimes), shifting the

composition of government revenue toward direct income taxation by **+1.76 pp** at +5 years and **+2.64 pp** at +10 years (nearly double the +0.90 and +1.53 pp shifts in low-investment regimes).

3. **Specificity Against Indirect Consumption Taxes:** On the negative-control channel (Δ Goods & Services Tax Share), high university investors show less than half the increase in indirect consumption tax reliance (+0.33 pp vs +0.70 pp at +5 years; +0.62 pp vs +1.31 pp at +10 years), confirming that the fiscal shift is specific to the direct income and corporate tax base rather than an across-the-board tax hike.